



Energy sharing in European renewable energy communities: Impact of regulated charges

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ARTICLE INFO

Handling editor: X Zhang

Keywords:

Collective self-consumption

Energy community

Electricity tariff

Fiscal policy

Local energy planning

ABSTRACT

The financial benefits of energy sharing within a community is impacted by factors within the control of the community, such as internal rules for energy sharing, and by external factors, such as regulatory frameworks and fiscal policies. While much of the literature on energy sharing has focused on internal rules, there has been less attention paid to the role of external factors, despite their potential policy implications. This study aims to address this gap by examining the range of fiscal measures that can be used to regulate energy sharing in renewable energy communities in 39 countries. The study considers two community arrangements: one in which prosumers share their surplus energy, and another in which consumers shares the energy generated by a collectively-owned PV. The findings suggest that the annual savings from energy sharing in the first arrangement may not be significant enough to justify the formation of the community. In contrast, the cost savings from sharing energy in the second arrangement are much greater, but also more influenced by the fiscal support policies. Based on the results, the study offers policy recommendations for avoiding resistance from stakeholders and aligning their incentives when introducing energy sharing regulation.

1. Introduction

As concerns about climate change and rising energy prices grow, households are increasingly interested in the benefits of investing in solar photovoltaic (PV) generation and sharing energy in communities [1]. This is particularly true in the European Union, where regulatory frameworks for energy sharing are starting to be established. There are two key aspects to energy sharing that are of interest to policy makers and researchers: the internal rules for distributing costs and benefits within the community, and the external regulations that establish the governing framework [2]. While both are important, the literature on energy sharing tends to focus more on the former than the latter.

Much of the research on energy sharing has focused on the internal rules by which energy is shared among community members [3]. A review of relevant models, solution concepts and algorithmic techniques can be found in Ref. [4]. With a theoretical basis grounded in game theory, this branch of the literature evolved from suggesting solutions such as the Shapley value [5], the nucleolus [6] and peer-to-peer pricing

mechanisms without uncertainty [7] and with uncertainty [8], to more recent approaches that suggest solutions based on so-called repartition keys that are better aligned with EU legislation. An example is a previous work by the authors [9] in which they propose the virtual net-billing method as a fair energy sharing method that is based on a dynamic ex-post repartition key. The benefit of using this method is assessed in terms of its fairness and scalability by comparing it to five other energy sharing methods from the literature, on a sample of 600 hypothetical communities. Other similar works which compare multiple repartition keys are those of Li et al. [10], which compares ten repartitions keys, Mustika et al. [11], which evaluates the interplay various repartition keys and energy management strategies and the work of Minuto et al. [12], which considers energy sharing in communities with different ownership structures. While these works provide guidance to energy community members on setting up fair energy sharing rules, they do not address the full range of measures needed to maximize the benefits of energy sharing.

In contrast, there has been considerably less research focused on the

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<https://doi.org/10.1016/j.energy.2023.128333>

Received 12 January 2023; Received in revised form 23 June 2023; Accepted 2 July 2023

Available online 3 July 2023

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impacts of different fiscal measures which outline the environment in which energy sharing takes place, leaving many questions unaddressed. Past research, such as [13,14], has studied how fiscal measures affect the economic benefits of individual prosumers. Along this line, we expand this line of research to energy communities and explore how the regulated charges (taxes, charges, fees and levies) applied to community members impact their benefits from sharing energy.

1.1. Related works

For energy consumers, fiscal policies are implemented through the design of electricity tariffs and the application of regulated charges [15]. Apart from ensuring the recovery of the infrastructure investment costs for the energy system, the tariff design should be simple, transparent and equitable [16], so as to effectively guide the investment and consumption behavior of energy consumers [17]. Having in mind that each tariff design comes with a compromise [18], it is worth exploring how the regulated charges should be included in the electricity tariffs so that they stimulate energy sharing in communities, instead of discouraging it.

Early research on the topic, which explored the allocation of network energy losses [19] and overall network charges [20], finds that network charges can be allocated in such a way that local energy exchanges are encouraged, but that this should be done equitably, in order to avoid discriminating against members in disadvantaged network locations. The works of Schreck et al. [21] and Maldet et al. [22], contribute to this knowledge by exploring the impact that different practical designs of network charges have on energy sharing. Specifically, Schreck et al. [21] compare four designs for the network charges applied in energy communities in Germany and find that charges based on the network connection capacity are more effective than volumetric changes in reducing peak power exchanges between the community and the distribution grid. Similarly, Maldet et al. [22] study how network charge designs affect the value of peer-to-peer energy trading, with a special focus on the regulation of Austria, Norway and Ireland. They also review the European regulation and conclude that the main barriers for implementing innovative network charges for energy communities stem from the trade-off between the economic benefits of community members and the ability of distribution system operators (DSOs) to fully recover their infrastructure investment costs.

However, the savings from energy sharing are impacted not only by the network charges, but also by the value added tax, renewable tax, capacity tax, environmental tax, nuclear taxes and others. Through a theoretical case study, Tsao et al. [23] illustrate the perspective of an energy company which determines the electricity price of prosumers that share energy, considering the payback time of their investments when they are sharing energy with other prosumers. The impact of the electricity price and its components are also studied by Radl et al. [24] which compare the profitability of energy sharing in eight European countries. Although they focus on optimal investment decision making, interesting insight can be drawn from their analysis, since each analyzed country differs in the composition of regulated charges in the retail price paid by consumers. From their analysis, the authors of [24] suggest that in the future, electricity tariffs should be redesigned following the polluter-pays principle and that further research is required to quantify the “sensitivity of the individual electricity price components” on the benefit of energy sharing.

Viti et al. [25] explore the impact of different regulated charge components of the electricity price and find that the savings from energy sharing in their analysis reduce from 12% to 9% when community members pay the network charges for the shared energy, and that the savings fall to 5% when both network charges and other regulated charges are paid. These costs savings are quantified against a scenario where the community members are individual prosumers and the analysis is relevant only for energy communities in Italy.

To the best of the authors knowledge, there is a lack of similar research which evaluates the impact of regulated charges on the benefit

of energy sharing for different European countries, despite the need for such analyses, given the reform of the electricity market design [26] and considering that many European countries are yet to fully develop their regulation on energy communities. This paper aims to address this research gap. A comprehensive overview of the different implementations of energy sharing regulation, both in the EU and worldwide, can be found in the works of Minuto et al. [12], Ines et al. [27], Frieden et al. [28,29] and Moura et al. [30]. Their findings show the convincing differences, both in terms of magnitude and implementation, of the support provided to energy communities. For instance, Austria and Portugal exempt communities from paying certain regulated charges, Italy and Germany incentivize communities to self-consume by paying them a premium for their self-consumption, Greece and Belgium apply implicit incentives by using virtual net-metering, while other countries have their regulation under development and are still evaluating how different configurations of energy sharing policies will support or hinder energy communities.

1.2. Contribution

The purpose of this paper is to explore the range of fiscal measures that policy makers can implement when establishing country-level regulations for energy sharing. We aim to quantify the impact of regulated charges (tax, charge, fee and levy) on the economic benefits of energy sharing across Europe. This research builds on previous work by offering a broader geographical perspective, studying 39 countries, and a broader conceptual perspective, examining all regulated charges. Through this study, we address the following research questions:

1. To what extent do different regulated charges applied to shared energy affect the cost savings in a community?
2. What is the impact of each regulated charge on community cost savings from energy sharing?
3. Given the post-pandemic energy crisis, how does increasing the energy and supply cost component of the electricity tariff affect cost savings?

We answer these questions for two arrangements of energy sharing, shown in Fig. 1. The first arrangement involves prosumers sharing their surplus generation among themselves, while the second represents a community of consumers sharing the generation of a collectively-owned PV system. To make the two arrangement comparable, the analyzed community has equal electricity generation and demand in the two arrangements.

2. Methods

This section presents the applied methods in the analysis of energy sharing in both energy sharing arrangements.

2.1. Configuring the energy billing in energy communities

In the first arrangement, each member of the energy community is described by their electricity load $p_{li}(t)$ and generation $p_{gi}(t)$ profile. Part of the electricity generation of member i is locally self-consumed $p_{sc,i}(t)$, while the rest is exported in the grid $p_{ex,i}(t)$.

Energy sharing in the community takes place when the electricity exports of some community members are instantaneously imported by other members in the community, where $p_{im,i}(t)$ denotes the electricity import of member i , where i is the index of the community member. The total shared electricity in the community in that case is calculated as follows:

$$p_{sh,N}(t) = \min \left(\sum_{i \in N} p_{ex,i}(t), \sum_{i \in N} p_{im,i}(t) \right) \quad (1)$$

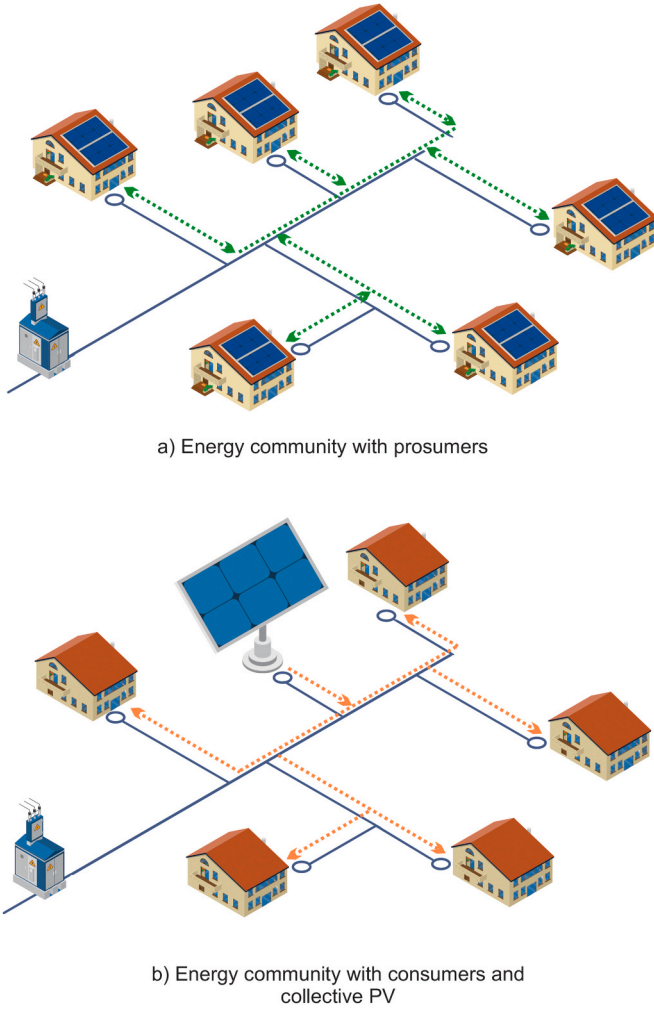


Fig. 1. Illustration of the two energy sharing arrangements analyzed in this paper.

If member i is not part of the community and instead acts as a single prosumer, their electricity bill, assuming flat tariffs for simplicity, is calculated based only on the electricity imports and exports:

$$C_{P,i} = E_{im,i}(c_{im} + c_n + c_t + c_v) - E_{ex,i}c_{ex} \quad (2)$$

where $E_{im,i}$ and $E_{ex,i}$ are the imported and exported electricity for a given billing period, c_{im} is the energy and supply price paid for the imported electricity, c_n is the network charge, c_t denotes other taxes (environmental, renewable, nuclear etc.), c_v is the value added tax and c_{ex} is price of the exported electricity, usually determined contractually between the supplier and prosumer such that $c_{ex} \leq c_{im}$. Given that the imported electricity is equal to the total load minus the self-consumed electricity $E_{im,i} = E_{l,i} - E_{sc,i}$, the previous equation can be rewritten as:

$$C_{P,i} = E_{l,i}(c_{im} + c_n + c_t + c_v) - E_{sc,i}(c_{im} + c_n + c_t + c_v) - E_{ex,i}c_{ex} \quad (3)$$

If part of a community, member i can share (exchange) energy with other community members by importing electricity while other members are exporting. As a result, member i virtually self-consumes some amount of energy $E_{sh,i}$ for which he/she pays a price c_{sh} to the other members of the community, such that $c_{ex} \leq c_{sh} \leq c_{im}$. Based on [9], the electricity bill of member i in this case is equal to:

$$C_{EC,i}^i = E_{l,i}(c_{im} + c_n + c_t + c_v) - E_{sc,i}(c_{im} + c_n + c_t + c_v) - E_{ex,i}c_{ex} - E_{sh,i}(c_{im} - c_{sh} + \sum_{e \in E} c_e) \quad (4)$$

The full derivation of (4) is given in the Supplementary Materials. Compared to the case when member i acts does not share energy, the cost savings from sharing energy are equal to $E_{sh,i}(c_{im} - c_{sh} + \sum_{e \in E} c_e)$. Clearly, they are proportional to the price difference between importing and sharing energy ($c_{im} - c_{sh}$) and are also proportional to the regulated charges $\sum_{e \in E} c_e$ from which the shared energy of member i is exempt. In this way, the financial support given to energy communities is seen through the lens of exemption of regulated charges. This analytical approach is general and adaptable, since it treats the regulated charges as a proxy for the incentive provided to energy communities without introducing additional subsidies. Once the amount of regulated charges and their impact of energy sharing is known, each country may choose how the incentive is put in place. For instance, it can be implemented in the form of a direct reimbursement (incentive), a tax relief, or any other form that is suitable considering the national framework conditions.

The same rationale transfers to the second arrangement, comprised of consumers sharing energy from a collective PV. The electricity bill of member i is reduced only due to the energy sharing, i.e. virtually self-consumption of the collective PV generation:

$$C_{EC,i}^i = E_{l,i}(c_{im} + c_n + c_t + c_v) - E_{sh,i} \left(c_{im} - c_{sh} + \sum_{e \in E} c_e \right) - E_{ex,i}c_{ex} \quad (5)$$

in this arrangement, $E_{sh,i}$ denotes the energy that is shared between the collective PV and the community member i . At the community level, the energy shared in this manner is, in absolute terms, equal to the locally self-consumed and shared energy from the previous arrangement, given that the capacity of the PV systems in the two arrangements are equal, thus making the two arrangements comparable. An illustration of the energy balances of the two arrangements is given in Fig. 2. As per (5), the benefit from energy sharing in the second arrangement depends on the shared energy, the price difference ($c_{im} - c_{sh}$) and the exemption of regulated charges $\sum_{e \in E} c_e$.

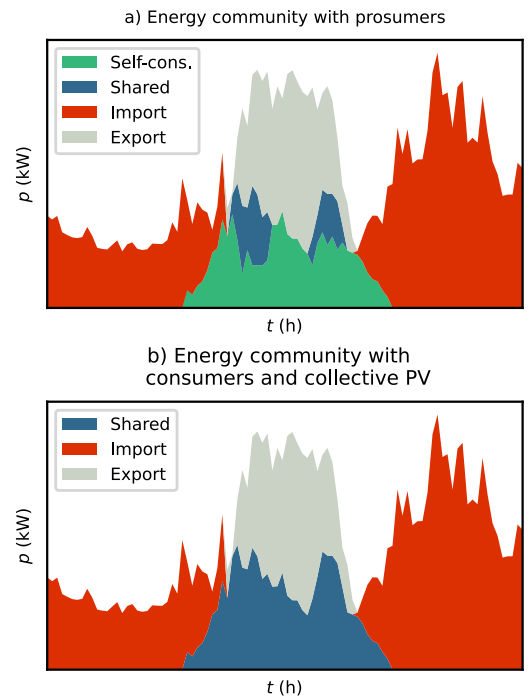


Fig. 2. Illustrative daily energy balance for an energy community in each energy sharing arrangement.

2.2. Distributing the shared electricity among the community members

In both arrangement, the electricity sharing among the community members is settled using a dynamic, ex-post repartition key, meaning that the settlement for the electricity sharing is performed for each time step of the billing period Δt , after the billing period ends. Then, the shared electricity is calculated as:

$$E_{sh,i} = \sum_{t \in T} p_{sh,i}(t) \Delta t \quad (6)$$

in this paper, the repartition key proposed in Ref. [9] is used, which underlies the virtual net-billing (VNB) method. According to this method, at each time step t , member i is allocated $p_{sh,i}(t)$ which represents a portion of the total shared electricity in the community $p_{sh,N}(t)$:

$$p_{sh,i}(t) = \alpha_i(t) p_{sh,N}(t) \quad (7)$$

The amount $p_{sh,i}(t)$ allocated to member i is based on the repartition key $\alpha_i(t)$:

$$\alpha_i(t) = \begin{cases} \frac{1}{2} \frac{p_{ex,i}(t)}{\sum_{i \in N} p_{ex,i}(t)}, & p_{ex,i}(t) > 0 \\ \frac{1}{2} \frac{p_{im,i}(t)}{\sum_{i \in N} p_{im,i}(t)}, & p_{im,i}(t) > 0 \\ 0, & \text{else} \end{cases} \quad (8)$$

Using (6)–(8), community members can determine how the collectively shared energy is dynamically distributed among them. Then, using (4) or (5), for each arrangement respectively, this quantify is deduced from the on-site metered consumption, while ensuring that no one in the community is left worse-off and that there is a cost-reflective payment of regulated charges. The repartition key $\alpha_i(t)$ guarantees that the total benefit of the community is fully distributed among the members and each member receives a payoff based on their contribution.

3. A case study of European renewable energy communities

To assess the impact of regulated charges on the benefits of energy sharing, we study the energy sharing in 39 countries, including EU Member States, members of the European Free Trade Association, and EU candidate countries. The analysis is conducted for energy communities which are given no additional incentives or subsidies, apart from the potential exemption of regulated charges. Each country has its own unique solar potential and electricity prices. To compare the benefits of energy sharing across countries, we adopt a single energy community with fixed electricity consumption and installed photovoltaic (PV) capacity and analyze its savings from energy sharing over the course of one year. The community consists of 50 households with a PV generation capacity of 100 kWp, similar to previous studies such as [12,31].

Two different energy sharing arrangements for this community are studied: in the first arrangement, the PV capacity is distributed among households on their rooftops, while in the second arrangement, the PV capacity is located elsewhere (e.g. on the rooftop of a building or on a nearby field). While there are other possible energy sharing arrangement, the two adopted arrangements seem to be most commonly discussed in the literature or applied in practice [2].

For each arrangement, an optimistic scenario (OS), in which the community is exempt from all regulated charges and a pessimistic scenario (PS), in which the community pays all regulated charges are analyzed. Furthermore, an analysis of the impact of different components of regulated charges is conducted. In the community, each member owns a portion of the total generation capacity proportional to their annual electricity demand.

The electricity consumption profiles of the community members are

adopted from Ref. [32]. The original dataset contains electricity measurements of 200 households from a field study in Germany with a 15-min temporal resolution. From this dataset, electricity consumption data for a period of one year of 50 households are selected. The selected households have an annual electricity consumption between 2500 and 5000 kWh. This range reflects the average electricity consumption of European households (3700 MWh) [33] and should be representative of a large share of households in most analyzed countries. For countries with average annual household electricity consumption outside of this range (e.g. 1700 kWh in Romania or 9000 kWh in Sweden), the results should still be valid, but representative for a smaller share of all households, i.e. representative of households which fit in the 2500–5000 kWh range. The hourly electricity consumption of the community for an average day of each month, which is shown in Fig. 3, illustrates that the load profile follows a common pattern for electricity consumption in households.

While the electricity consumption and the installed PV capacity of the community are held constant, the conditions in which the community operates vary depending on the country in which it is located. The first condition that affects the energy sharing in the community is the PV electricity generation. To calculate the energy produced by the PV when the energy community is in given country, PVGIS [34] is used. Hourly generation profiles for the PV are calculated for the capital city of each country, using the SARA-2 solar irradiation data, for the latest available year, assuming optimized azimuth and slope. The time step of the dataset is $\Delta t = 1$. As a result, the electricity consumption data are aligned with this time step and averaged to 1 h values.

The third set of input data used in the analysis represent the electricity price data. The electricity prices used in the analysis are taken from the Eurostat database, which contains the average electricity price components for household consumers for 2021, which at the time of the analysis was the latest available data [35]. In general, the retail electricity price paid by households contains components for the energy and supply c_{im} , network charge c_n , value added tax (VAT) c_v and other taxes denoted as c_t (equal to the sum of renewable tax, capacity tax, environmental tax, nuclear taxes and other regulated charges), as defined in Ref. [35]. The composition of the retail electricity price of households, based on this data, is shown in Fig. 4. Deductions and tax returns on electricity bills are omitted from the analysis as they are variable, and potentially transitory. For the energy sharing, c_{sh} and c_{ex} are assumed to be constant and equal to half of c_{im} , which should reflect the trend of lower wholesale prices during daytime and also stimulate self-consumption.

To achieve greater transparency and replicability of the results, all input data (consumption, generation and prices) are made available in the Supplementary materials.

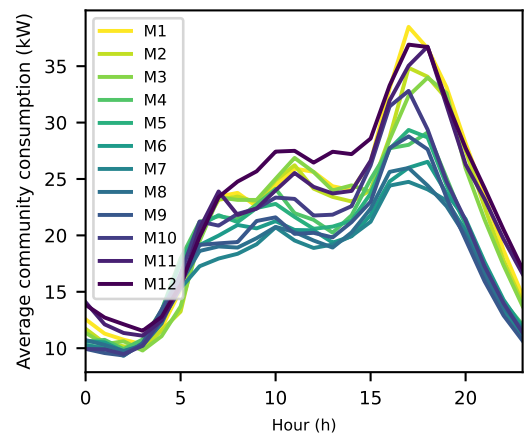


Fig. 3. Average daily electricity consumption of the studied energy community for each month.

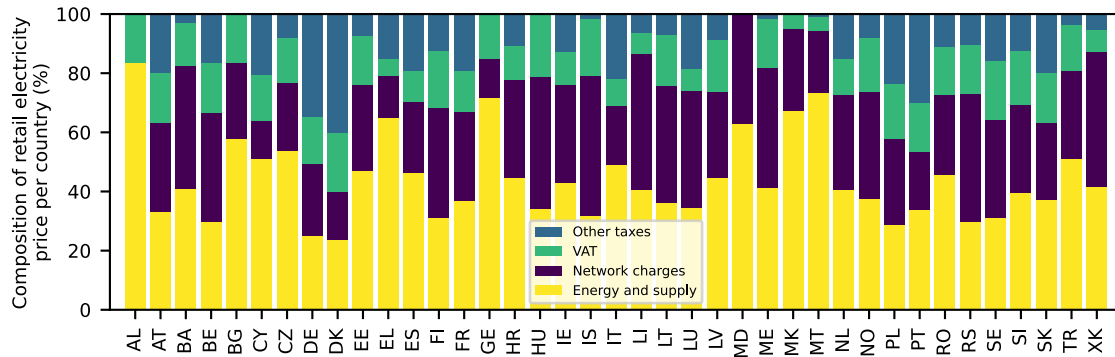


Fig. 4. Composition of household retail electricity price. Regulated charges are composed of the network charges, VAT and the other taxes.

4. Results

This section presents the results on the cost savings calculated for both energy sharing arrangements, as well as the impact of increasing energy and supply costs.

4.1. Energy sharing between prosumers

In the first energy sharing arrangement, community members locally self-consume a portion of their on-site generation and share their excess generation with the other members that need it at that point in time. The community members therefore benefit from two simultaneous forms of self-consumption: the physical self-consumption that occurs behind-the-meter, and the virtual self-consumption which occurs due to the energy sharing. Fig. 5 shows the relative reduction of electricity costs in the community, for a selected set of countries chosen to represent different geographical regions and different compositions of retail electricity price. In this figure, the yellow bar represents a business as usual (BaU) scenario, in which it is assumed that the community members have no PVs, the green bar (PV) shows the cost reduction obtained by local-self consumption of the rooftop PV energy, the blue bar represents the

relative cost reduction from energy sharing (ES) and the purple bar represents the remaining costs of the community in the OS, when all the regulated charges are avoided.

Depending on the country, the collective costs of the community are reduced by $(25.96 \pm 4.79)\%$ due to the local, behind-the-meter self-consumption. In comparison, energy sharing contributes to the savings on the collective bill considerably less, by about $(3.80 \pm 0.63)\%$. The community yields the biggest benefit from sharing energy when it is exposed to the conditions in Portugal, Cyprus and Italy, while it yields the lowest benefit when it is located in Iceland, Estonia and Latvia. Nevertheless, it is expected that the benefit from sharing energy decreases when regulated charges are introduced. Fig. 6 illustrates this point, showing how much the cost savings are influenced by each regulated charge. In the PS, when the community members pay all regulated charges, they benefit only because they share energy at a price c_{sh} , which is lower than the import price c_{im} . For the studied community, the cost reduction from this mechanism across the 39 countries is rather small and falls in the range $(1.11 \pm 0.54)\%$. It is only after the community is exempt from paying some of the regulated charges, that the benefit becomes more notable.

The network charge c_n exemption has the biggest influence on the

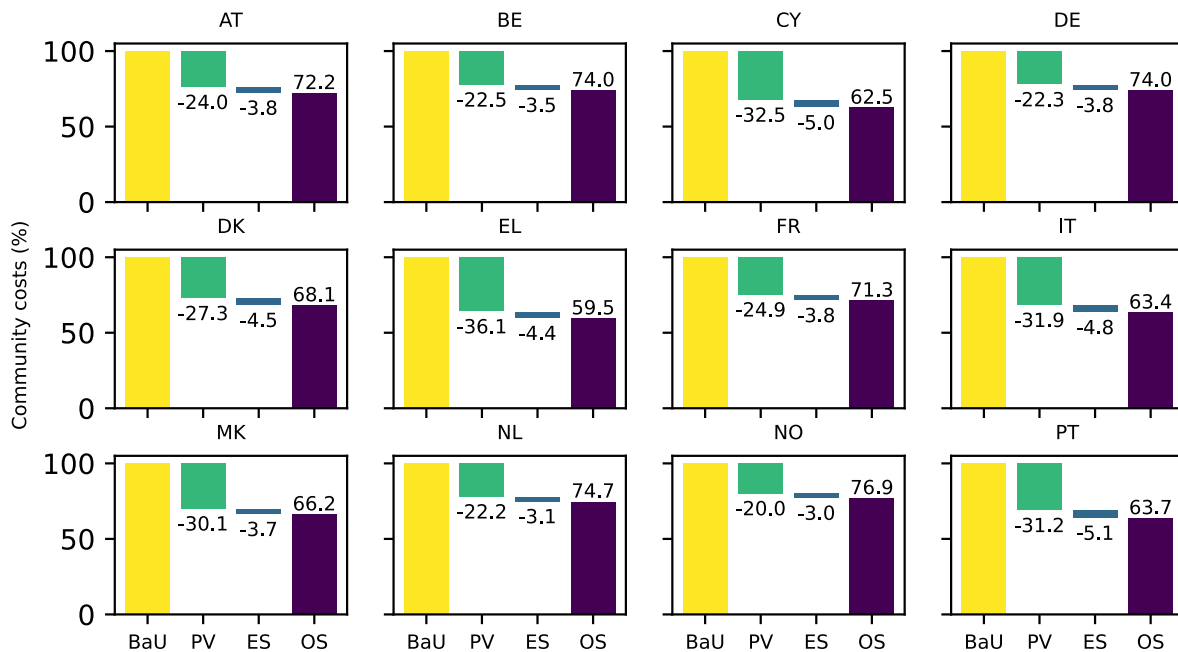


Fig. 5. Cost reduction of the community electricity bill in selected countries for the first arrangement (BaU: Relative community electricity costs when households have no PV and do not share energy; PV: Cost reduction from behind-the-meter self-consumption, when households are individual prosumers; ES: Cost reductions when households share energy; OS: Relative community cost when community members are exempt from all regulated charges).

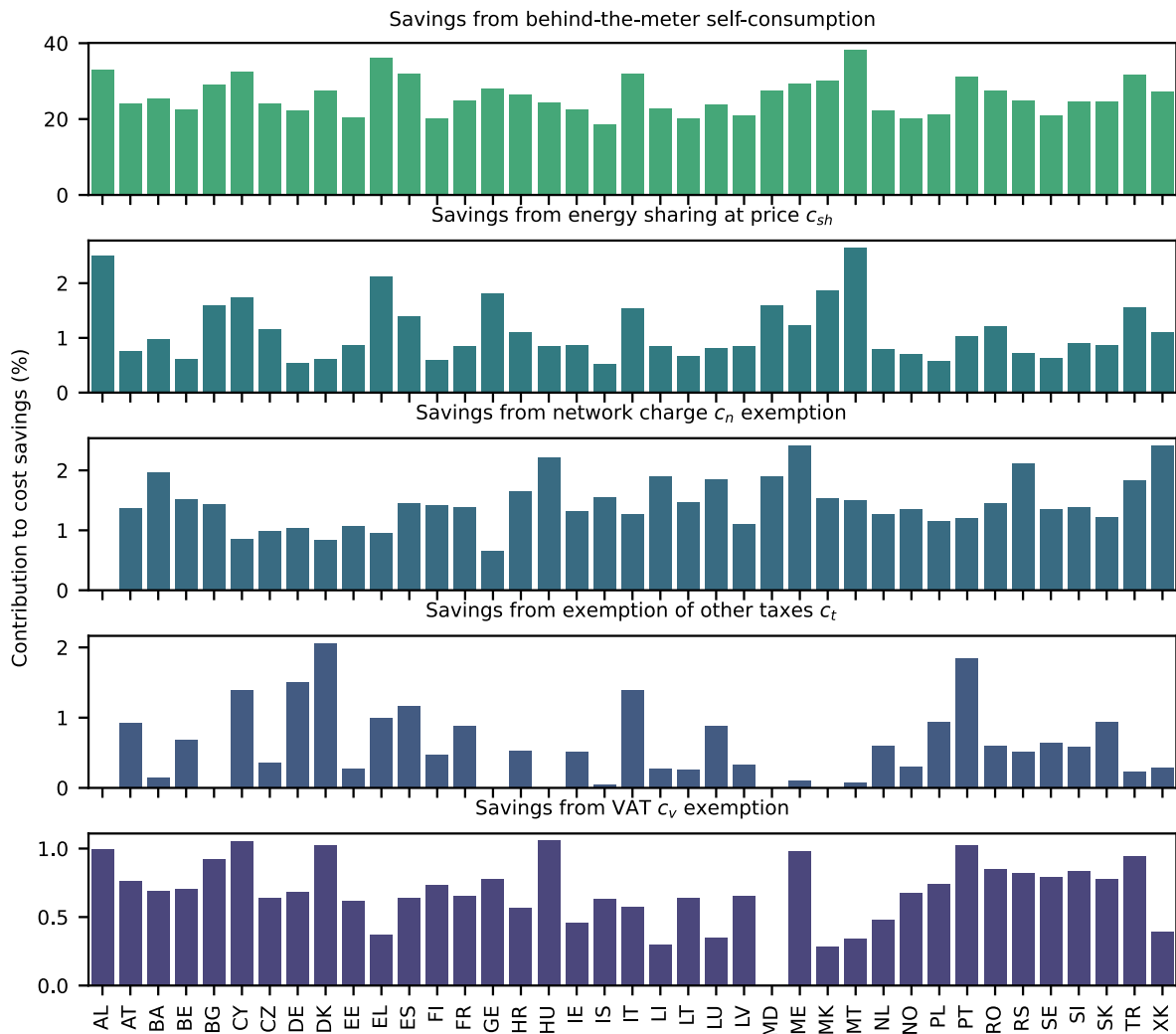


Fig. 6. Contribution to cost savings in comparison to BaU scenario for the first energy sharing arrangement.

cost savings, additionally reducing the collective bill of the community by $(1.42 \pm 0.47)\%$. The impact of other taxes c_t varies widely from country to country, depending on country-specific policies. When community members are relieved from paying these charges, their costs drop by $(0.59 \pm 0.53)\%$. The benefit from this exemption is most pronounced in Denmark, Germany, Cyprus and Portugal, where energy consumers pay relatively high regulated charges. At the same time, consumers in Albania, Bulgaria, Georgia, Hungary, Moldova and North Macedonia pay little to no such taxes and therefore do not benefit from this mechanism. Finally, if the community is exempt from paying the value added tax c_{vat} for the energy sharing, its collective costs reduce by an additional $(0.68 \pm 0.24)\%$.

4.2. Sharing the generation of a collective PV generator

In the second arrangement, when the prosumers share energy from a collective PV generator, the magnitude of their cost savings are larger, but are also more dependent on the fiscal environment. Unlike actual prosumers, the members of such a community do not physically reduce their energy imports from the distribution grid. Instead, when their electricity bill is calculated, it must account for the virtual self-consumption (energy sharing) calculated according to a pre-defined energy sharing method.

Fig. 7 shows the relative cost reduction of the community's bills in this arrangement for the same set of countries previously displayed in Fig. 5. The results show that the community can reduce its electricity bill

by $(24.61 \pm 3.69)\%$, compared to a BaU scenario without a PV, which is slightly lower than the combined cost savings from local self-consumption and energy sharing seen in the previous energy sharing arrangement.

These values also refer to the OS, under the assumption that the members of the community are exempt from the regulated charges for the energy that they virtually self-consume. Since the amount of energy that is shared in this arrangement is much greater than in the previous one, the application of regulated charges more notably impacts the costs savings. The relative contribution of each electricity price component on the costs savings in this case are shown in Fig. 8. From these results, one finds that when the regulated charges apply, the collective electricity costs of the community decrease by $(9.27 \pm 4.35)\%$. However, if the community is exempt from (i) network charges, (ii) other taxes or (iii) the value added tax, its electricity costs reduce by $(8.10 \pm 2.64)\%$, $(3.36 \pm 3.04)\%$ and $(3.86 \pm 1.34)\%$, respectively.

4.3. Impact of energy and supply costs

Abrupt surges of energy demand or constraints on the supply side may increase the wholesale prices in electricity markets. A combination of both of these effects are seen in many European countries in the post-pandemic period, leading to higher costs for energy and supply paid by final consumers. Since higher energy costs motivate households to invest in renewable energy sources for their own self-consumption, in this subsection, the impact of rising energy and supply costs on energy

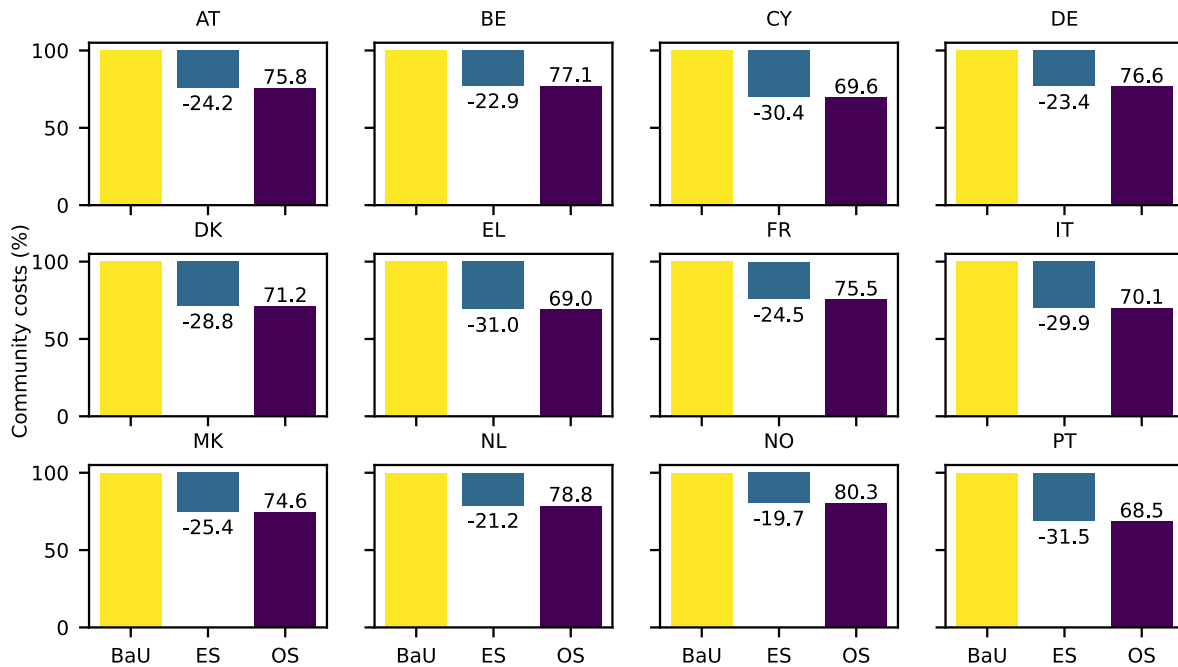


Fig. 7. Cost reduction of the community electricity bill in selected countries for the second arrangement (Description: BaU: Relative community costs for electricity when households have no PV and do not share energy; ES: Cost reductions when households share energy; OS: Relative community cost for electricity compared to BaU).

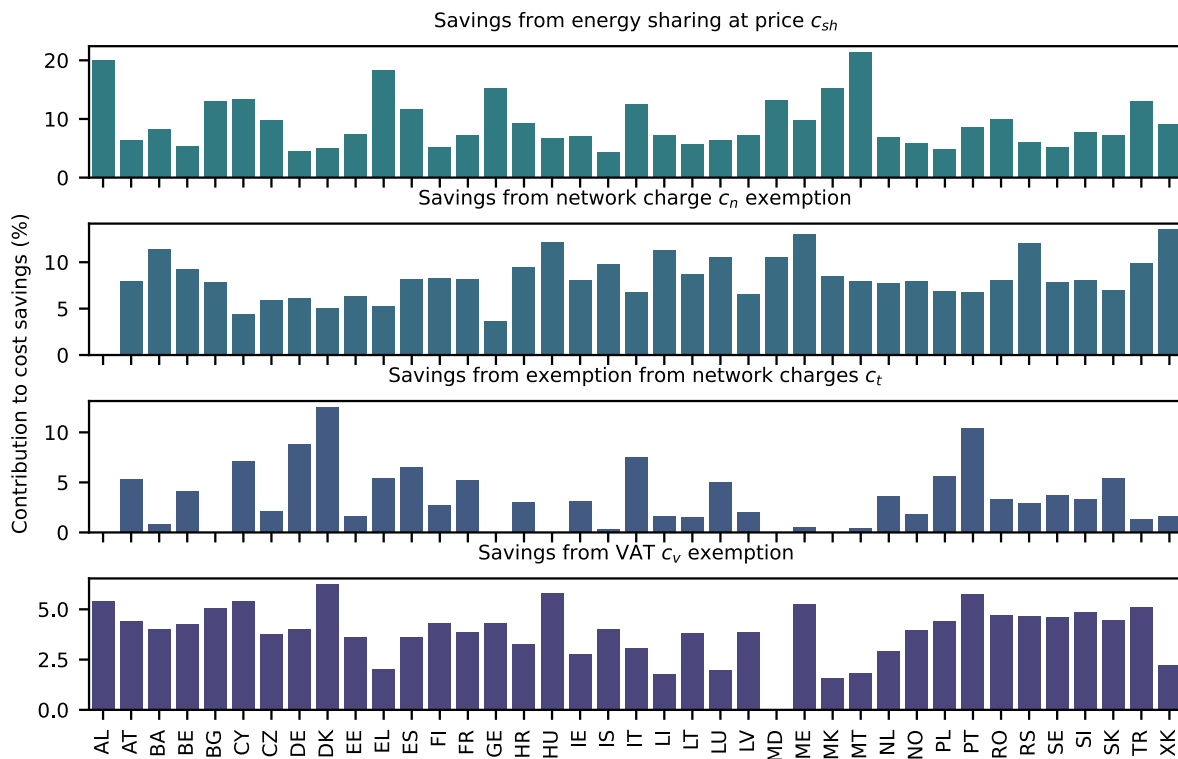


Fig. 8. Contribution to cost savings in comparison to BaU scenario in the second energy sharing arrangement.

sharing are analyzed. This is performed by repeating the analysis of the previous two subsections for five more cases in which the energy and supply costs are 2–6 times larger than those of the base year. Since equal cost multipliers are analyzed in all countries, the analysis should reveal general trends instead of country-specific findings.

The results for a community with the first energy sharing arrangement (prosumers sharing their excess generation), are given in Fig. 9.

The first subplot in the figure represents the analysis conducted for the base year, while the other subplots represent scenarios in which the energy and supply costs increase 2–6 times. On the y-axis, the figure shows the average annual cost savings of a community member, while on the x-axis it shows the average annual electricity bill of a community member in the BaU scenario. Each point in the plot represents a separate country. The yellow points represent the OS, while the blue points

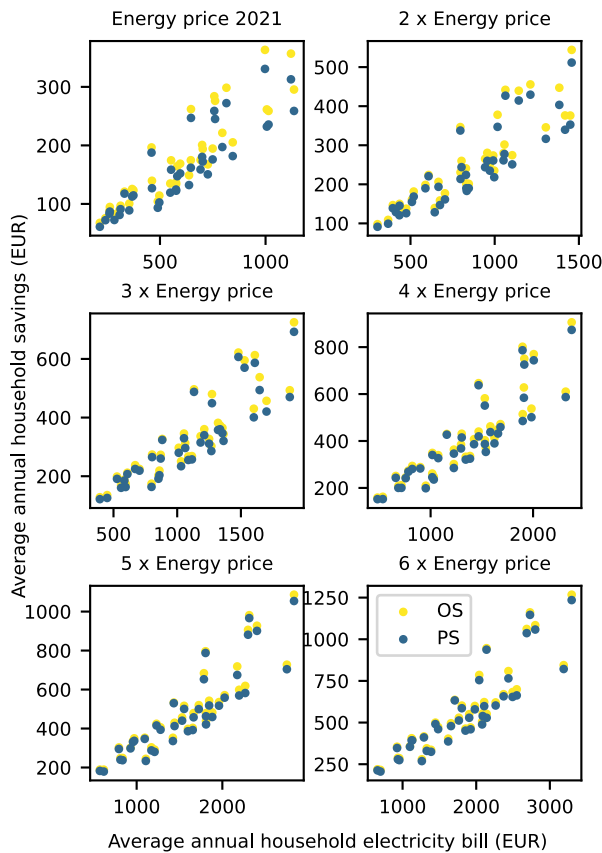


Fig. 9. Impact of energy and supply cost increase in first energy sharing arrangement: Average annual savings as a function of the average annual electricity costs in BaU.

represent a PS. Fig. 9 confirms that in the first energy sharing arrangement, applying regulated charges to the shared energy has a small impact on the overall costs savings. This is because the greatest costs savings in this arrangement come from local self-consumption, as illustrated in Figs. 2 and 6, and this quantity is not affected by the regulated charges applied to the shared energy. In Fig. 9, the marginal impact of the regulated charges is shown by the small differences between the yellow and the blue points. However, Fig. 9 also shows that as energy and supply cost increase, the benefit from avoiding regulated charges becomes smaller. Instead, the main savings come from sharing energy at a lower price than c_{im} . This is illustrated in Fig. 9 by the closing gap between the yellow and blue points for higher energy and supply costs.

Fig. 10 illustrates the results obtained when the same analytical procedure is applied to the second energy sharing arrangement. It shows that the differences between the savings in the OS and PS are much larger in the second arrangement compared to the first arrangement. The reason is that the amount of energy that is shared is much greater in the second arrangement, as shown in Fig. 2. As a result, the impact of the regulated charges on the cost savings are much more notable – they reduce the savings by 62.98% on average, or between 21.11% and 82.58% depending on the country. While following the PS may mitigate the revenue losses of DSOs and governments, it will disincentive the formation of energy communities and, as a result, could lead to lower investments in distributed energy sources and a missed opportunity to protect consumers against volatile energy prices [26].

Nevertheless, Fig. 10 shows that as the energy and supply costs increase, the difference between the savings in the OS and PS become much smaller, which reaffirms the conclusions drawn from Fig. 9. The reason for this is that the benefit comes from the avoidance of high energy and supply costs c_{im} by buying energy at a lower price c_{sh} from

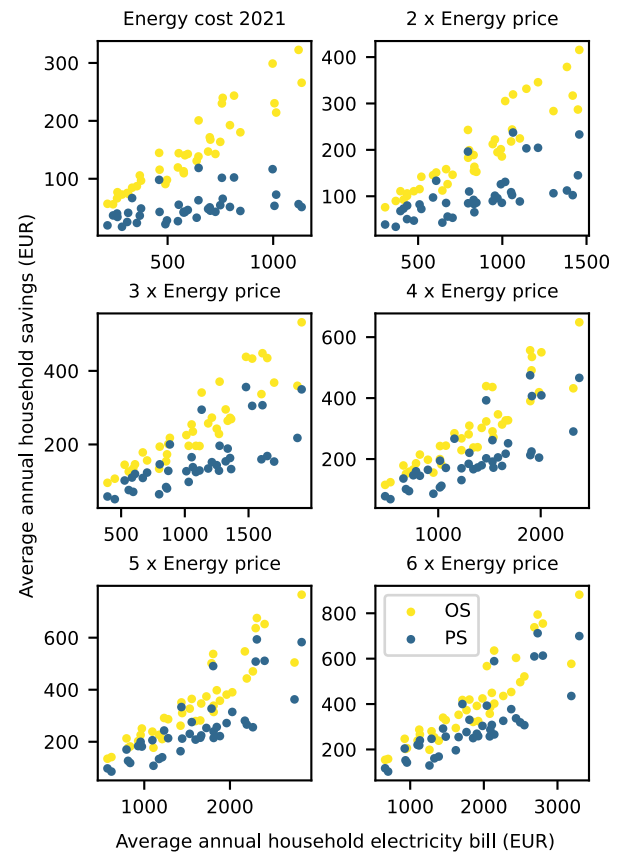


Fig. 10. Impact of energy and supply cost increase in second energy sharing arrangement: Average annual savings as a function of the average annual electricity costs in BaU.

the collective generator which can be owned, for instance, by a cooperative. As the energy and supply costs increase, the payment of regulated charges become tolerable since they have only a small impact on the overall cost savings. A potentially interesting line of research would be to investigate the point of energy and supply cost increase at which paying the regulated charges would not affect the profitability of investing in a community energy system.

4.4. Relationship between cost savings, solar irradiation and regulated charges

When the relative costs savings of the community are visualized on a geographical map, some interesting patterns emerge, as shown in Fig. 11. The first plot refers to the first arrangement, while the second refers to the second arrangement. The figure shows that the community tends to have greater savings when located in countries with lower latitudes, i.e. higher solar irradiation. However, there are outliers to this trend, such as Denmark, where the costs savings comes due to the exemption of regulated charges, which constitute about 2/3 of the retail electricity price of consumers. To better understand this relationship, the relative costs savings are plotted as a function of (i) the share of regulated charges in the retail electricity price and (ii) the specific PV energy generation (kWh/kWp) of each country. The results are shown in Fig. 12, which should be read as follows – each circle in the figure represents a different country.

The x-axis value of a circle represents the share of regulated charges in the retail electricity price in that country, while the y-axis value of a circle represents the specific PV energy generation in the capital city of that country. Finally, the color of the circle represents the relative costs savings from energy sharing, compared to a scenario without energy sharing. The values shown in this figure are equal to the blue colored

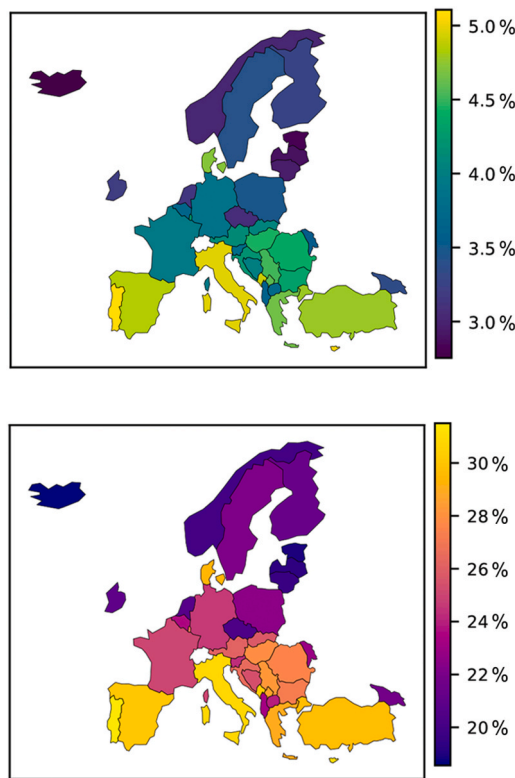


Fig. 11. Collective cost savings of the community from energy sharing in OS; Upper figure - first arrangement; Lower figure - second arrangement.

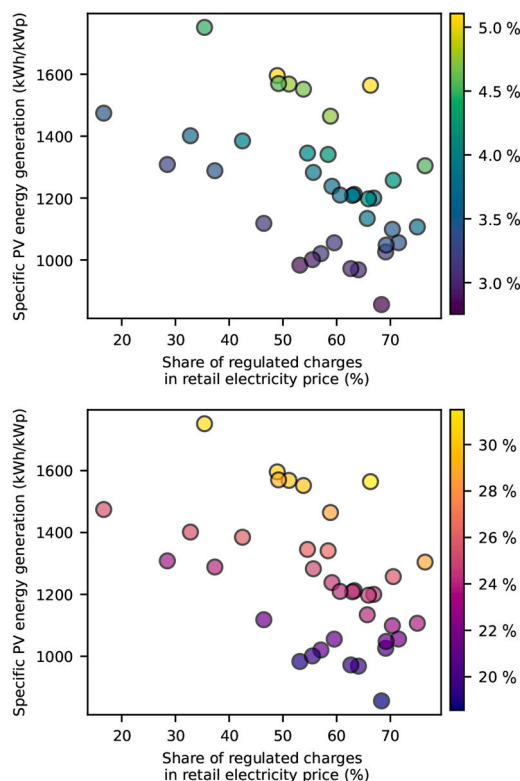


Fig. 12. Dependence of collective cost savings from energy sharing in OS on solar potential and regulated charges; Upper figure – first arrangement; Lower figure – second arrangement.

bars in Figs. 5 and 7. The results indicate that solar irradiation is not the only factor determining cost savings, and that there are countries in regions with high solar potential that have lower savings than their neighbors, and vice versa. The relative savings seem approximately constant over isolines which are slightly inclined from the upper left area towards the lower right area of the figure. This illustrates that the savings may be proportional to both factor, meaning that largest savings are obtained by communities in areas with high solar potential, which are also exempt from paying relatively high regulated charges. All results obtained from the analysis are provided as a supplementary file to this paper.

5. Discussion

The European Union Directive 2018/2001 [36] establishes rules for renewable energy communities, stating that members should be able to freely share energy according to internally defined rules and be subject to “cost-reflective network charges, as well as relevant charges, levies, and taxes” that ensure their fair contribution to total systems costs. However, policy makers have found that implementing these guidelines is challenging [26], as it requires aligning incentives among different actors, balancing economic and administrative burden, and finding the appropriate timing.

As a result, the fiscal policies and support mechanisms across Europe vary widely. For example, instead of exemption from all regulated charges, Italian energy communities are exempt from paying only part of the network charge (around 9 EUR/MWh), but are directly reimbursed 110 EUR/MWh for the self-consumed electricity. Nevertheless, effect of this policy is as if community members are exempt from paying about all regulated charges. In Austria, energy communities with members in the same distribution feeder pay a reduced network charge, since they only use a part of the electricity grid. In Germany, the owner of the collective PV generator in multi-apartment buildings receives a premium of 23.7–37.9 EUR/MWh for the self-consumed electricity, depending on the capacity of the PV generator, similarly to Italy. Moreover, the tenants of the building which use the electricity from the collective PV avoid some of the regulated charges. A more detailed overview of existing policies and support schemes across Europe can be found in Refs. [37,38].

Given the diversity of these policies and the uncertainty of their effects, in this paper we investigated the impact of such policies on renewable energy communities and explored whether regulated charges paid by community members affect the amount of savings derived from energy sharing. Our analysis confirms that this is the case, but also reveals a number of nuanced outcomes.

5.1. Quantifying the benefit of energy sharing

The findings from the comparison of the two energy sharing arrangements reveal that the potential monetary savings for households can vary significantly. As presented in subsection 4.1, the cost savings in the first arrangement are about $(3.80 \pm 0.63)\%$, in the OS. In the first arrangement, by sharing energy for one full year, a household saves less than a *monthly* electricity bill. More precisely, in the OS it saves about $(62.32 \pm 13.93)\%$ of a monthly electricity bill, while in the PS, when all charges are paid, the *annual* savings drop to $(18.65 \pm 10.48)\%$ of a monthly electricity bill.

In absolute values, the average annual cost savings only due to sharing energy, per community member, range from 5.88 to 29.67 EUR in Austria, 4.97–17.46 EUR in Croatia, 5.97–41.83 EUR in Germany, 13.41–28.11 EUR in Greece, and 5.12–28.19 EUR in Sweden, among others. These findings suggest that when the community is composed solely of prosumers which own the same type of generation source, which is common for households with PVs, there is a lack of diversity in the net-load profiles of the community members, leading to small amounts of energy being shared. Hence, energy sharing provides only a

marginal benefit in this arrangement, which is much smaller than the benefit of the local self-consumption. While specific to the community examined, the results suggest that the potential savings from energy sharing in such groups may not be significant on an annual basis. These findings enhance our understanding of the issue obtained through previous research [39]. Moreover, they raise questions whether the benefit of this energy sharing arrangement would be sufficient to offset any costs associated with administration or management of the community's energy sharing system [40].

With that in mind, it would be useful to consider two alternatives of this energy sharing arrangement that could provide greater value to community members and other stakeholders. The first alternative would be for the community members to own some flexible assets, such as energy storage systems, electric vehicles, or power-to-X elements [41], which link different energy sectors within the community [42]. Through smart management of these assets, members of a community will be able to shift their energy consumption to better align it with the excess generation of other community members, thus increasing their savings. In doing so, along with increasing their self-consumption, they may also potentially provide added value to other stakeholders (e.g. to the DSOs by mitigating overvoltages or line overloads) [43]. Another alternative arrangement is the formation of communities with members that naturally have diverse load profiles, such as PV prosumers, households, small commercial consumers, electric vehicle charging stations etc. [44].

In comparison for the first arrangement, the community cost savings from sharing energy in the second arrangement are much greater, since considerably more energy is shared. An example of a community in this arrangement is a multi-apartment building with collective rooftop PV, where energy is shared behind the point of common coupling to the distribution network [45]. For multi-apartment buildings it is reasonable to relieve the community from paying network charges, as in Austria and Germany. In the OS, depending on the country, the average annual cost savings from sharing energy in this arrangement are equal to about three monthly electricity bills (2.95 ± 0.44). This would be equal to about 188.76 EUR in Austria, 133.37 EUR in Croatia, 260.60 EUR in Germany, 186.85 EUR in Greece, and 176.71 EUR in Sweden. Conversely, if the community members and the collective PV generator are located at different points in the distribution network, they may be subject to paying a full or reduced distribution network charge. If regulation requires it, community members may even be required to pay all regulated charges, as in the PS. Depending on the country, the costs savings in the PS are (62.98 ± 14.29)% lower than in the OS.

This may be detrimental to the profitability of investing in a community energy system. While purchasing a collective PV generator of a certain capacity usually costs less than purchasing multiple small PVs with the same capacity, community members in the second arrangement are usually required to have some legal entity, such as a cooperative. Maintaining a legal entity, such as a cooperative, additionally reduces the cost savings of energy sharing, since the members incur administrative costs. Hence, the savings of community members from sharing energy and the investment savings from purchasing a collective PV, instead of individual rooftop PVs, should counterbalance the administrative costs of maintaining a legal entity for an energy community and the possible regulated charges paid for the shared energy, in order for the investment to be profitable.

5.2. Timing regulatory changes

One notable policy recommendation from our study is related to the timing of bringing forth regulatory change related to energy sharing. Our results suggest that regulatory changes may be more easily implemented and more beneficial for all parties when energy and supply costs are high. As illustrated in Figs. 9 and 10, when energy and supply costs are 4–6 times higher than they were in 2021, the economic impact on community members of paying regulated charges for shared energy is minimal. This presents an opportunity for policy makers in Europe to

introduce energy sharing regulations with minimal resistance from stakeholders. During times of high wholesale prices, energy consumers are more interested in investing in renewable energy for their self-consumption, since it enables them to significantly reduce their electricity bills. When this is done in an energy community, it may not be necessary to fully exempt community members from paying taxes, fees, and other charges for the energy they share in order to incentivize such behavior. The reason for this, as mentioned previously, is because community members in this case benefit from avoiding the high energy and supply costs, not from avoiding regulated charges. Instead, the main challenge will be to ensure that energy is consumed and shared within the community rather than sold on the market. If successful, energy sharing under these circumstances can align the incentives of energy consumers seeking to reduce their bills with those of policy makers seeking to implement new regulations, governments seeking to meet energy and climate targets, and distribution system operators looking to avoid revenue losses and avoid new investment.

5.3. Limitations and further research ideas

This study has several limitations that should be noted. One limitation is the use of fixed electricity consumption, which does not respond to PV generation or energy sharing. While this was done to explore the worst-case benefits that can be obtained when there is no flexibility on the demand side, a natural extension of this work would be to include flexible assets such as electric vehicles and heat pumps. An analysis including heat pumps would be particularly interesting, considering the impact of ambient temperature on heating and cooling demand and the coefficient of performance of the heat pumps.

Another limitation of this study is that it only examines fiscal measures implemented through electricity tariff design. There are other measures that can significantly support energy communities, such as subsidizing investments in collective infrastructure (such as PV systems, storage, heat pumps, or electric vehicle charging infrastructure) or offering technical support and reduced administrative procedures to community members. Future research could, for example, examine the impact of offering such support for investments in infrastructure that can also benefit other stakeholders (e.g. smart charging of electric vehicles or batteries to avoid overvoltage issues during peak PV generation for distribution system operators).

Furthermore, the analysis utilizes price data from the year 2021 and represents a snapshot of a single year. However, it should be noted that the fiscal measurements of countries can change rapidly. In this case, while the results of this study should hold qualitatively, any potential changes in the fiscal measures and the regulated charges are expected to influence the quantitative output of the analysis. A useful avenue for future research would be to extend the analysis from a single year to the lifetime of the community energy system, to evaluate profitability considering geographical differences in investment and capital costs, as well as the differences between investing in collective PV generators and individual rooftop PV generation units. With inflation, it is expected that the gap between collective and individual investments could widen.

6. Conclusion

This study has examined the impact of the regulated charges on energy sharing in European renewable energy communities. The results show that the potential savings for households from energy sharing can vary significantly, depending on the specific arrangement and the choice of fiscal support mechanisms. When the community is comprised only of households with PVs, without any regulated charges, a household can save approximately 62.32% of their monthly electricity bill over the course of one year through energy sharing in a community. However, when all regulated charges are taken into account, the annual savings drop to 18.65% of a monthly electricity bill. These findings suggest that the potential savings from energy sharing in this arrangement may not

be significant on an annual basis and may not be sufficient to offset the costs associated with managing the community. However, introducing flexible assets or including consumers next to the prosumers in the community can increase the value of energy sharing in this arrangement.

In comparison, the community cost savings from sharing energy in the second arrangement are about 5 times greater and equivalent to nearly three monthly electricity bills, since considerably more energy is shared. These types of communities are possible in multi-apartment buildings or in suburban/rural areas, where collectively owned generation units can be located nearby the community members. It is worth noting that in this arrangement, although community members benefit more from sharing energy in comparison to prosumers share energy, they yield lower savings overall (since they have not local self-consumption) and their savings are *more* dependent on the charges, taxes, fees and levies applied to the shared energy.

The study also found that distributing the fiscal support among stakeholders, such as distribution network operators and governments, can be a challenge when implementing energy sharing regulations. In order to avoid resistance from these stakeholders, it was suggested to introduce energy sharing regulation when energy and supply costs are high, since during this period community member can highly benefit from self-consumption and energy sharing, even if they pay most regulated charges. Another suggestion for avoiding resistance is to move from a zero-sum framework to a superadditive one, by introducing measures that offer co-benefits, such as subsidies in demand side flexibility, so that communities can better balance local supply and demand and thus avoid overvoltage and overloading issues at the distribution level.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2023.128333>.

Appendix: abbreviations and symbols

Table of abbreviations

DSO	Distribution system operator
EU	European Union
NB	Net-billing
OS	Optimistic scenario
PS	Pessimistic scenario
PV	Photovoltaic
SV	Shapley value
VAT	Value added tax
VNB	Virtual net-billing

Table of symbols

Parameter	Definition
i	Index of community member
N	Set of prosumers in community
E	Set of potentially exemptible regulated charges
n	Maximum number of prosumers in community
T	Set of time-steps in analyzed time horizon
t	Time-step
Δt	Duration of time-step t
$p_{l,i}(t)$	Electricity consumption of prosumer i
$p_{g,i}(t)$	Local electricity generation of prosumer i
$p_{m,i}(t)$	Electricity import of member i at the point of common coupling
$p_{ex,i}(t)$	Electricity exports of member i at the point of common coupling
$p_{sc,i}(t)$	Individual self-consumption by member i , due to overlap in local load and generation
$p_{sh,N}(t)$	Shared electricity in energy community N
$p_{sh,i}(t)$	Portion of the shared electricity $p_{sh,N}(t)$ allocated to member i
$E_{l,i}$	Energy demand of member i
$E_{m,i}$	Imported energy by member i

(continued on next page)

CRediT authorship contribution statement

Vladimir Z. Gjorgievski: Investigation, Software, Methodology, Formal analysis, Writing – original draft, Funding acquisition. **Bodan Velkovski:** Investigation, Software, Formal analysis, Writing – review & editing. **Minuto Francesco Demetrio:** Conceptualization, Formal analysis, Writing – review & editing. **Snezana Cundeva:** Methodology, Supervision, Writing – review & editing. **Natasa Markovska:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Data availability

Data will be made available on request.

Acknowledgement

This work has been financed by the Ss. Cyril and Methodius University in Skopje under NIP. UKIM.22–23.12. The authors would also like to acknowledge Blagoja Markovski for the fruitful discussion and suggestions on the manuscript. The authors would also like to thank the anonymous reviewers for the constructive comments.

(continued)

Parameter	Definition
$E_{ex,i}$	Exported energy by member i
$E_{sh,i}$	Portion of the total shared energy allocated to member i
$E_{sc,i}$	Self-consumed energy (behind-the-meter) by community members i with a rooftop PV generator
$C_{p,i}$	Electricity cost of member i calculated based on individual net-billing (applicable only for community members with rooftop PVs)
$C'_{EC,i}$	Electricity cost of prosumer i calculated based on VNB in energy sharing arrangement 1
$C''_{EC,i}$	Electricity cost of prosumer i calculated based on VNB in energy sharing arrangement 2
c_{im}	Electricity tariff for the net-imported electricity
c_{ex}	Electricity tariff for the net-exported electricity
c_{sh}	Electricity tariff associated with the energy shared in the community
c_n	Network charges
c_t	Other taxes
c_v	Value added tax
$\alpha_i(t)$	Proportional contribution of member i to the excess self-consumption

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